

Assignment 5

A- Functions with Arrays Tasks

1. Print All Elements of an Array Using a Function

Write a program that defines a function to print all elements of an array. The function should take the array as a parameter and print its elements.

Example:

Input: Array elements = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Output: Array elements = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

2. Find the Sum of Array Elements Using a Function

Write a program that defines a function to calculate the sum of all elements in an array. The function should take the array as a parameter and return the sum.

Example:

Input: Array elements = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Output: Sum of array = 55

3. Find the Sum of Main Diagonal Elements of a Matrix

Write a function that takes a square matrix as a parameter and returns the sum of the elements on the main diagonal.

Example:

Input: Matrix =

1 2 3

4 5 6

7 8 9

Output: 15

B- String Tasks

4. Find the Most Repeated Character in a String

Write a function that takes a string as a parameter and returns the character that occurs most frequently. If there is a tie (**i.e. two letters with the same number of occurrences**), return any one of the most repeated characters.

Example:

Input: String = aaabca

Output: a

5. Convert a String to Lowercase

Write a function that takes a string as a parameter and converts all characters to lowercase. The function should return the resulting string.

Example:

Input: String = AbCD

Output: abcd

6. Check if All Characters in a String Are Digits

Write a function that takes a string as a parameter and checks if all characters in the string are digits. The function should return **true** if all characters are digits, otherwise return **false**.

Examples:

Input: String = 12354

Output: true

Input: String = a123

Output: false

7. Reverse an Integer Using Recursion

Write a recursive function that takes an integer as a parameter and returns its reverse.

Example:

Input: Integer = 5324

Output: 4235

8. Find the Maximum Digit in an Integer

Write a function that takes an integer as a parameter and returns the largest digit in the number.

Example:

Input: Integer = 5234

Output: 5

9. Reverse a String Using a Loop

Write a function that takes a string as a parameter and reverse it using a loop.

Example:

Input: String = Hello

Output: olleH

10. Count the Total Number of Words in a String

Write a program to count the total number of words in a given string. Words are separated by spaces.

Example:

Input: String = I love Code.

Output: Total number of words = 3

C- Pointers Tasks

11. Write a function that takes an array of integers of size 10 and returns an array with only the even numbers.

Example:

Input: {4, 7, 3, 9, 2, 13, 6, 7, 8, 11}

Output: {4, 2, 6, 8}

12. Write a function to swap two numbers using pointers and display their values before and after swapping in the main function.

13. Write a program to dynamically allocate memory for an integer array, read values from the user, and print them.

Example:

Input: 1 9 2 1 5

Output: 1 9 2 1 5

14. Find the output of the following code without using Visual Studio or Code::Blocks:

```
#include <iostream>
using namespace std;
int main() {
    int arr[] = {4, 5, 6, 7};
    int *p = (arr + 1);
    cout << *arr + 9;
    return 0;
}
```

Options:

- a) 12
- b) 14
- c) 13
- d) Error

15. What will happen in the following code without using Visual Studio or Code::Blocks?

```
int a = 100, b = 200;  
int *p = &a, *q = &b;  
p = q;
```

Options:

- a) b is assigned to a
 - b) p now points to b
 - c) a is assigned to b
 - d) q now points to a
-

Bonus Tasks

1. Change Each Letter to the Next Letter in the Alphabet

Write a C++ program to change every letter in a given string to the letter following it in the alphabet. The letter 'z' should wrap around to 'a'. Non-alphabetic characters should remain unchanged.

(i.e. a becomes b, p becomes q, z becomes a, etc.)

Example:

Input: String = w3resource

Output: x3sftpv sdf

2. Capitalize the First Letter of Each Word

Write a C++ program to capitalize the first letter of each word in a given string. **Words are separated by a single space.**

Example:

Input: String = cpp string exercises

Output: Cpp String Exercises

3. Remove Anchor (#) from a URL

Write a function to return a URL with anything after the anchor symbol ('#') removed.

Examples:

Input: URL = 'www.code.com#about'

Output: 'www.code.com'

Input: URL = 'www.code.com?page=1'

Output: 'www.code.com?page=1'